

STUDENT CASE STUDY EXECUTION REPORT

DEPARTMENT	FED
SUBJECT	DSA-2
ACADEMIC YEAR	I-III Semester (2025-26)
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COURSE OUTCOME	

- **Case Study Topic:** FoodChain Backend Restaurant & Delivery Network Optimization using BFS, DFS, Kruskal's MST, and Prim's MST

➤ **Work Done Summary:**

In this case study, we implemented a graph-based FoodChain Backend Restaurant & Delivery Index Management System to optimize delivery infrastructure and network connectivity.

The delivery network was modeled as a weighted undirected graph where restaurants, warehouses, and delivery hubs were represented as vertices and road connections as weighted edges.

We applied **BFS** to locate the nearest delivery hubs and analyze service coverage areas. **DFS** was used to verify network connectivity and identify clusters of restaurants. We constructed **Minimum Spanning Trees (MSTs)** using **Kruskal's** and **Prim's algorithms** to minimize infrastructure costs and design efficient delivery routes.

The implementation demonstrated how graph traversal and MST algorithms can improve logistics planning and operational efficiency in large-scale food delivery systems.

➤ **Implementation Details**

1. Represented restaurants, warehouses, and delivery hubs as vertices in a weighted undirected graph.
2. Stored road distances and travel costs as edge weights between locations.
3. Applied **Breadth First Search (BFS)** to find the nearest available delivery hub and analyze reachable service areas.
4. Applied **Depth First Search (DFS)** to verify network connectivity and discover regional restaurant clusters.
5. Implemented **Kruskal's Algorithm** using Disjoint Set (Union-Find) to construct a minimum-cost infrastructure network.
6. Implemented **Prim's Algorithm** using a Priority Queue to expand delivery routes from a central warehouse.
7. Compared the performance of BFS, DFS, Kruskal's, and Prim's algorithms for delivery network optimization.
8. Concluded that MST-based solutions significantly reduce infrastructure costs, while BFS and DFS efficiently support routing and connectivity analysis with time complexity **$O(V + E)$** .

➤ RESULT: Screenshot/Output

```
PS C:\Users\Sai\OneDrive\Desktop\DSA case study 3> java Main
>>
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FOODCHAIN DELIVERY NETWORK MANAGEMENT
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Location Mapping:
0 - Central Warehouse
1 - Restaurant A
2 - Restaurant B
3 - Delivery Hub X
4 - Delivery Hub Y
5 - Restaurant C

BFS Traversal:
0 1 2 3 4 5

DFS Traversal:
0 1 3 2 4 5

Connected Components:
Cluster 1: 0 1 3 2 4 5
Total Clusters = 1

Kruskal MST:
3 -- 4 = 1 km
1 -- 3 = 2 km
0 -- 2 = 3 km
0 -- 1 = 4 km
4 -- 5 = 7 km
Total MST Cost = 17 km

Prim MST:
0 -- 2 = 3 km
0 -- 1 = 4 km
1 -- 3 = 2 km
3 -- 4 = 1 km
4 -- 5 = 7 km
Total MST Cost = 17 km
PS C:\Users\Sai\OneDrive\Desktop\DSA case study 3>
```

The BFS, DFS, Connected Components, Kruskal & Prim MST, Union-Find.....etc.

Top 5 recent timestamps returned were:

101, 250, 55, 300, 150

Time Complexity:

- BFS, DFS, Connected Components - $O(V + E)$
- Kruskal MST & Prim MST – $O(E \log E)$ & $O(E \log V)$
- Union-Find - $O(\alpha(V)) \approx O(1)$

The **FoodChain Backend Restaurant & Delivery Network Management System** successfully applied graph-based Data Structures and Algorithms to solve real-world logistics problems. By using **BFS** and **DFS**, the system efficiently analyzed delivery routes, service coverage, and network connectivity. **Kruskal's** and **Prim's Minimum Spanning Tree algorithms** were used to design a cost-effective delivery infrastructure while minimizing overall road construction and maintenance costs. The project demonstrated the practical use of **graphs, graph traversal, connected components, priority queues, and union-find data structures**, providing an optimized and scalable solution for restaurant and delivery network management.

GitHub link: <https://github.com/slashnight/DSA-2-case-study-assignment>

Student Signature	Faculty Signature